Yuanchao Xu

xu.yuanchao.3a@kyoto-u.ac.jp

Room 4-405, Maskawa Building for Education and Research, Kyoto University

June 22, 2026

Working Experience

University of Alberta
Visiting Postdoc, Applied Mathematics
April 2026 - August 2026

Kyoto University
Postdoc, SACRA
September 2025 - August 2030

Education Experience

University of Alberta
Doctor of Philosophy (*Dissertation Award*), Applied Mathematics
September 2021 - August 2025

University of Manitoba
Master of Science, Mathematics
September 2018 - August 2020

University of Oregon
Bachelor of Science (*Cum Laude*), Mathematics
January 2015 - December 2017

Research Interest

My research interest lies in data-driven methods for stochastic dynamical system. In particular, I am interested in building up theoretical framework for learning stochastic Koopman operator and applying it into complex real-world problems, e.g., computational neuroscience, reinforcement learning and generative modeling, etc.

Research

1. Generative Modeling through Koopman Spectral Analysis: An Operator-Theoretic Perspective (arXiv, 2025)
2. Reinforcement-learning-guided data-driven estimation of spectral properties of stochastic Koopman semigroups (DCDS-I, 2025)
3. Spectral analysis of Koopman operator through pseudo-resolvent ((Submitted to Physica D) arXiv, 2025)
4. A Data-Driven Framework for Koopman Semigroup Estimation in Stochastic Dynamical Systems (Chaos, 2025)
5. Koopman Eigenfunctions Links Multiscale State-Dependent Brain Dynamics (In preperation, 2025)
6. ResKoopNet: Learning Koopman Representations for Complex Dynamics with Spectral Residuals (ICML2025, Vancouver, Canada, 2025)
7. Koopman Spectral Analysis Uncovers the Temporal Structure of Spontaneous Neural Events (COSYNE (poster), Lisbon, Portugal, 2024)
8. Decentralized Multi-Agent Reinforcement Learning for Task Offloading Under Uncertainty (arXiv, 2021)

Talk

1. *Generative Modeling through Koopman Spectral Analysis*
   Massachusetts Institute of Technology, Cambridge, MA, April 27, 2026
2. *Generative Modeling through Koopman Spectral Analysis*
   CMS Special Seminar
   California Institute of Technology, Pasadena, CA, April 13, 2026
3. *Generative Modeling through Koopman Spectral Analysis*
   Probability Seminar
   University of Oregon, Eugene, OR, April 8, 2026

4. *Generative Modeling through Koopman Spectral Analysis*
   MLDS 5 & DEDS 2026
   Kyoto University, Kyoto, Japan, February 9-13, 2026
5. *Generative Modeling through Koopman Spectral Analysis*
   Machine Learning and Dynamical Systems Seminar
   Alan Turing Institute, London, UK, January 8, 2026 (Online)
6. *Data-Driven Koopman Framework for Modeling Complex Dynamical Systems*
   Fudan University, Shanghai, China, January 5, 2026
7. *Data-Driven Koopman Framework for Modeling Complex Dynamical Systems*
   European Molecular Biology Laboratory, Rome, Italy, November 24, 2025 (Online)
8. *Data-Driven Koopman Framework for Modeling Complex Dynamical Systems*
   Laboratoire Hubert Curien, Saint-Étienne, France, October 14, 2025
9. *Perturbation method for learning stochastic Koopman operator*
   CREST
   RIKEN, Kobe, Japan, December 26-28, 2024
10. *Extracting Dynamics from Complex Systems: Deterministic and Stochastic Perspectives*
    East China University of Science and Technology, Shanghai, China, December 19, 2024

Visiting

1. Massachusetts Institute of Technology, Cambridge, MA, April 27-30
2. Cornell University, Ithaca, NY, April 16-18
3. California Institute of Technology, Pasadena, CA, April 13-15
4. University of California, Santa Barbara, CA, April 10th
5. University of Oregon, Eugene, OR, April 6-9
6. Fudan University, Shanghai, China, January 5-9, 2026
7. Laboratoire Hubert Curien, Saint-Étienne, France, October 14-15, 2025
8. European Molecular Biology Laboratory, Rome, Italy, October 11-13, 2025
9. Istituto Italiano di Tecnologia, Genoa, Italy, October 9-10, 2025
10. Ehime University, Matsuyama, Japan, December 23-25, 2024

Reviewer

ICML2026, ICLR2026, IEEE-TNNLS, IEEE-TETCI

Service

Young Associate Editor: DCDS-I

Awards

- Faculty of Science Doctoral Dissertation Award, 2025
- Mary Louise Imrie Graduate Student Award, 2025
- Dr. Josephine M. Mitchell Recruitment Scholarship, 2021
- International Graduate Student Entrance Scholarship, 2020
- International Graduate Student Entrance Scholarship, 2018
- Clarence and Lucille Dunbar Scholarship, 2017

Teaching Experience

Math 102(lab) Applied Linear Algebra	*Winter 2025*
Math 209(lab) Calculus for Engineering 3	*Fall 2024*
Math 209(lab) Calculus for Engineering 3	*Spring 2024*
Math 102(lab) Applied Linear Algebra	*Winter 2024*
Math 201(lab) Differential Equations	*Fall 2023*
Math 101(lab) Calculus for Engineering 2	*Spring 2023*
Math 201(lab) Differential Equations	*Winter 2023*
Graduate Research Assistant	*September 2020 - August 2021*
Graduate Teaching Assistant	*September 2018 - August 2020*
Undergraduate Teaching Assistant	*September 2016 - December 2017*

Undergraduate Math Tutor *April 2016 - December 2017*